

Improved Offset Handling in Hand-centered Object Manipulation Extending Ray-casting

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1 INTRODUCTION

As Virtual Reality (VR) is becoming a readily available tool in an increasing number of fields, we see also an increasing number of non-technical users entering virtual environments (VEs). One pose manipulation technique that has the potential to be both fast and intuitive to use, is the Hand-centered Object Manipulation Extending Ray-casting (HOMER), presented first by Bowman and Hodges [1]. In this paper we take a look at how a naive implementation of HOMER can lead to unintuitive interaction and present mathematics for avoiding this problem, by using a rotational offset of the grab vector instead of a static offset vector. Subsequently we compare the naive and rotation-based versions with respect to task completion time and number of re-grabs, and measure their respective System Usability Scale (SUS) score and Raw NASA Task Load Index (RTLX)

2 IMPROVING MANIPULATION CONTROL

With HOMER (principles illustrated in Fig. 1) the user selects an object using a ray extending from their wand and activates manipulation e.g. by the press of a button, see Fig. 1(a). A distance ratio, $r_{\text{dist}} = |\vec{p}_{\text{grab}} - \vec{p}_{\text{wand}}| / |\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}|$, is recorded at the moment of grabbing which is then subsequently used, during pose manipulation, to set the object's position,

$$\vec{p}_{\text{grab}} = \vec{p}_{\text{wand}} + r_{\text{dist}}(\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}) \quad (1)$$

This allows the user to move the object, apart from sideways, also closer and further away, see Fig. 1(b).

2.1 Offset-based HOMER

A problem with the original version of HOMER stem from the fact that the object position is determined by the user-wand line; in cases where the object, wand and user are not lining up, an undesirable, potentially large, spatial jump can result, as illustrated in Fig. 1. To remove this artifact, the manipulation-time grab position must initially co-inside with the selection-time grab position, as illustrated in Fig. 2. One such solution [2] includes an object offset adjustment, $\vec{v}_{\text{offset}}$ in Fig. 2(a), placing the object at

$$\vec{p}_{\text{grab}} = \vec{p}_{\text{wand}} + r_{\text{dist}}(\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}) + \vec{v}_{\text{offset}} \quad (2)$$

The offset, $\vec{v}_{\text{offset}}$, is estimated using the orthogonal projection of the intersection point onto the user-wand line,

$$\vec{p}'_{\text{grab}} = \vec{p}_{\text{user}} + (\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}) \frac{\vec{p}_{\text{grab}} \cdot (\vec{p}_{\text{wand}} - \vec{p}_{\text{user}})}{|\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}|^2} \quad (3)$$

$$\vec{v}_{\text{offset}} = \vec{p}_{\text{grab}} - \vec{p}'_{\text{grab}} \quad (4)$$

and to get this right also r_{dist} should now be estimated using the projection of the grab point, $\vec{p}'_{\text{grab}}$, instead of $\vec{p}_{\text{grab}}$. It should also be

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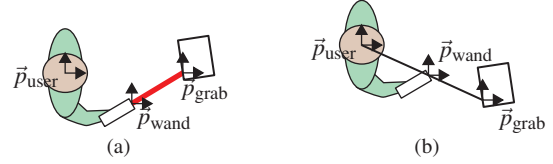


Figure 1: Original HOMER, illustrating a) selection and then b) grabbing, resulting in an unwanted jump in position.

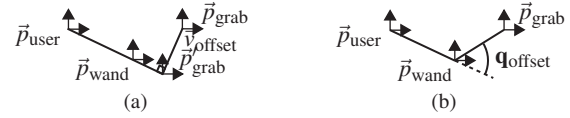


Figure 2: Solutions to avoid the unwanted jump with a) positional offset, $\vec{v}_{\text{offset}}$, and b) rotational offset, $\mathbf{q}_{\text{offset}}$, when grabbing the object.

mentioned that Wilkes and Bowman during manipulation multiplied the offset vector with $|\vec{p}_{\text{wand}} - \vec{p}_{\text{user}}| / |\vec{p}_{\text{wand}}^0 - \vec{p}_{\text{user}}^0|$ where $\vec{p}_{\text{wand}}^0$ and $\vec{p}_{\text{user}}^0$ are the wand and user positions at initialization of the manipulation, respectively. This scales the length of the offset vector together with the translation towards and away from the user.

Implementing this, however, we were unable to find details on how to handle the orientation of this offset in 3D space, for example if the user turn around to face another direction. We suspect that this rotational inconsistency can have an impact on the natural interaction by novice users.

2.2 Rotation-based HOMER

To avoid a static offset, we propose the use of a rotational offset from the user-wand vector, as illustrated in Fig. 2(b). This can be implemented in different ways, with different representations of rotation, but here we use quaternion representation of rotation. Assuming that every vector $\vec{v} \in \mathbb{R}^3$ can be represented by a pure quaternion $\mathbf{v} \in \mathbb{R}^4$, thus allowing a unit quaternion $\mathbf{q}$ to rotate a vector $\vec{v}$ through $\vec{v}' = \mathbf{q}\vec{v}\mathbf{q}^*$, where $\mathbf{q}^*$ denotes the conjugate of $\mathbf{q}$, we can express the HOMER translation as,

$$\vec{p}_{\text{grab}} = \vec{p}_{\text{wand}} + r_{\text{dist}}\mathbf{q}_{\text{offset}}(\vec{p}_{\text{wand}} - \vec{p}_{\text{user}})\mathbf{q}_{\text{offset}}^* \quad (5)$$

where $\mathbf{q}_{\text{offset}}$ is a quaternion encoding the rotational offset.

Finding this offset is a matter of finding the quaternion that encodes the rotation of the user-wand line into the wand-grab line, see Fig. 2(b). A unit quaternion representing a rotation has the values $\mathbf{q} = [\cos \frac{\phi}{2}, \hat{r} \sin \frac{\phi}{2}]$, where $\hat{r}$ and ϕ are the axis and angle of the represented rotation. Also, the angle and axis between two vectors $\vec{a}$ and $\vec{c}$ is given by $\vec{a} \cdot \vec{c} = |\vec{a}| \cdot |\vec{c}| \cdot \cos \phi$ and $\hat{r} = \frac{\vec{a} \times \vec{c}}{|\vec{a} \times \vec{c}|}$, respectively. From this it is straightforward to calculate the offset quaternion, or

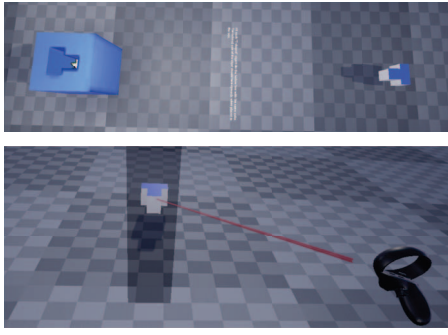


Figure 3: Test scene with object (top right) and hole (top left) and horizontal obstacle (bottom).

we can even put the equations together into

$$\mathbf{q}_{\text{offset}} = \left[\sqrt{\frac{1}{2} + \frac{\vec{a} \cdot \vec{c}}{2 \cdot |\vec{a}| \cdot |\vec{c}|}}, \frac{\vec{a} \times \vec{c}}{|\vec{a} \times \vec{c}|} \cdot \sqrt{\frac{1}{2} - \frac{\vec{a} \cdot \vec{c}}{2 \cdot |\vec{a}| \cdot |\vec{c}|}} \right] \quad (6)$$

where $\vec{a} = \vec{p}_{\text{wand}} - \vec{p}_{\text{user}}$ and $\vec{c} = \vec{p}_{\text{grab}} - \vec{p}_{\text{wand}}$.

2.3 Anchor Point Location

In a pilot test with the HOMER technique, we found that the experience can be improved by optimizing the location of the “user” point, $\vec{p}_{\text{user}}$. After testing key egocentric locations (head, chest, belly and shoulder), we found that using the shoulder joint of the working hand provides the best ergonomics and precision in translations both sideways and distance-wise. This anchor point was used for both versions in the evaluation below.

3 EVALUATION

A total of 12 people, 3 female and 9 male, participated in a study of the two approaches, all with little experience with VR equipment and applications. The reported self-assessed mean experience in VR for all subjects was 1.83, ranked from 1–5 (from *never used* to *very experienced*). The youngest participant was 20 years old and the oldest 68 years old, the average age was 32 and the median age 24.

After an introduction to the test the subject performed a learning session to familiarize themselves with the equipment (Oculus Quest 2), the VE and the two versions of HOMER. The task was then to pick up a “T”-shaped object, turn around to face the opposite direction and then place the object, colored side up, in a “T”-shaped hole, slightly larger than the object, see Fig. 3 (top). After each correct drop a new object appeared with initial orientation randomly chosen from a predefined set. This randomness, combined with the shape of the object, forces the subject to apply different rotation to the object to correctly align it with the hole. Each subject had to place a total of twelve object in the box to complete the test procedure. To simulate an untidy environment without adding an unpredictable randomness to the test setup, each object appeared either behind a vertical obstacle or a horizontal obstacle, or freely accessible, see Fig. 3 (bottom).

Each subject performed the procedure twice, once with each HOMER version, in balanced order. For each version, time-to-finish each task and the number of re-grabs were recorded, and the subject filled in the SUS and NASA TLX forms.

4 RESULTS

The mean (and std dev) of completion time was 192.8 s (84.1) and 161.0 s (55.8) for the offset-based and rotation-based techniques, respectively. This difference is statistically significant, $p \approx 0.0214$. The mean (and std dev) number of re-grabs was 9.8 (4.5) and 8.3

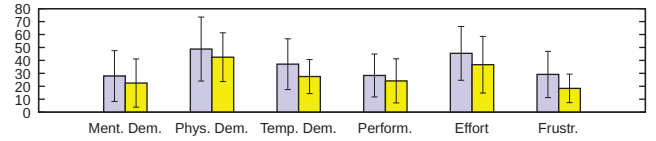
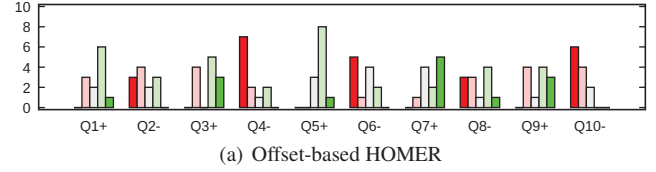
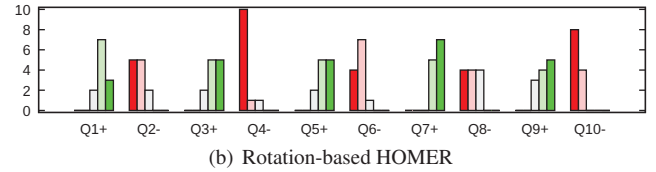


Figure 4: Mean and standard deviation of RTLX for offset- (blue, left) and rotation-based (yellow, right) HOMER.



(a) Offset-based HOMER



(b) Rotation-based HOMER

Figure 5: Histogram over the number of subjects providing each answer, in five steps from strongly disagree (red, leftmost) to strongly agree (green, rightmost), to each question in the SUS form, here also marked with ‘+’ for agreeing is better or ‘-’ for disagreeing is better.

(5.5) for the offset-based and rotation-based techniques, respectively. This difference is *not* statistically significant, $p \approx 0.219$.

The mean overall RTLX was 36.1 for the offset-based implementation and 28.6 for the rotation-based. With Shapiro-Wilk showing no significant departure from the normality, t-test was used to show that the difference is statistically significant with $p \approx 2.1 \cdot 10^{-4}$. RTLX also provides an index for each of six categories, the statistics of which are shown in Fig. 4. Each question in the SUS form is answered with how much, in a five level Likert scale, the subject agrees to its statement and our results are therefore shown in histogram form, in Fig. 5. A total score can also be calculated, resulting in 69.0 for the offset-based HOMER and 82.7 for the rotation-based. Again allowed by Shapiro-Wilk, t-test was used to show that the is significant difference with $p \approx 0.004$.

5 CONCLUSIONS

In this paper we have described a way to avoid the rotational inconsistencies of a naive implementation of the HOMER interaction technique. Also, we have reported on indications that HOMER interaction works best if its “user point” is positioned at the shoulder for the hand used for manipulation. We have also presented a comparison between HOMER with static offset vector and with rotational offset, using 12 subjects and four different measures: completion time, number of re-grabs, SUS score and RTLX, showing improvements on almost all measures with the proposed approach.

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